Software Requirements Specification

Brain Training
Bookstore Online Purchase and Home Delivery Management System
Estándar: IEEE Std. 830-1998
Fecha: October 11, 2026

Prepared by team CtrlZ: Andres Daquilema, Lino Diaz, Mateo Dominguez

1. Introduction

1.1. Purpose

The purpose of this document is to define the specifications, features, non-functional requirements, and constraints of the system. The system will be used by:

1. Software Development Team: For designing the system, the book database, and any modules that are created along the way.
2. Bookstore Staff: To keep the system running properly.
3. Delivery Person: To manage the shipping system properly.

1.2. System Scope

The software is officially named “Brain Training BookStore”.

What the system will do:

- Customer Module:
    - Register and Log in
    - Browse book catalog, search for books, and view book details
    - Manage shopping cart
    - Place an order and pay for an order
    - View orders and check delivery status
- Admin Module:
    - Register and Log in
    - View orders
    - Manage books and manage inventory
    - Manage deliveries
    - Sales reports and account management
- Delivery Driver Module:
    - Log in
    - View assigned deliveries and view delivery address details
    - Update delivery status

What the system will not do:

1. It will not manage employee salaries.
2. It will not track the condition of the delivery vehicles.
3. It won't be responsible for all of the bookstore's accounting, but it will prepare basic sales reports.
4. It will not manage book orders from suppliers.

Objectives of system:

1. Increase sales
2. Keep the inventory up to date and cut down on errors
3. Reach more customers online
4. Improve the user experience when shopping online

1.3. Definitions, Acronyms, and Abbrevations

SRS	Software Requirements Specification
IEEE	Institute of Electrical and Electronics Engineers
Module	A part of the system that groups related functions.

1.4. References

IEEE, *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Std 830-1998, IEEE Computer Society, 1998.

1.5. Document Overview

This document has three sections. The first one, presented above, explains the purpose of the system, what it will do, what it will not do, and its objectives. Section 2 gives an overall description of the product, users, and constraints. Section 3 details the specific requirements for system realization.

2. General Description

The proposed system consists of a web platform for managing purchases and deliveries for a bookstore located in Quito, Ecuador. Its purpose is to make it easier for customers to browse the book catalog, select products, and place orders online, while also allowing them to track the status of their purchases throughout the preparation and delivery process.

The platform aims to enhance the shopping experience through an accessible and intuitive interface, which displays relevant information about the books—such as titles, authors, prices, and availability. It also includes an order-tracking feature, enabling customers to monitor the progress of their purchases from the moment of confirmation until the products are received.

The system is designed to optimize the interaction between the bookstore and its customers by centralizing product and order information within a single digital environment. In this way, it seeks to streamline the purchasing process, improve the organization of deliveries, and provide

greater transparency regarding the status of orders.

2.1. Product Perspective

The system will be an independent web application designed to complement the bookstore's commercial activities by digitizing the process of browsing, purchasing, and delivering books.

The platform will serve as a communication channel between customers and the bookstore, allowing users to view available products and place orders without needing to visit the physical store to consult the catalog.

Conceptually, the system will consist of a digital catalog, a shopping cart management module, an order placement process, and delivery tracking. These components will be interconnected to provide a seamless shopping experience, from the selection of books to their final delivery.

The solution may evolve in the future to incorporate integrations with external services—such as electronic payment platforms or geolocation tools—should business needs require it.

2.2. Product Functions

In general terms, the system will provide the following functions:

- Catalog browsing: allows users to view books registered on the platform and see their key details.

- Book search: facilitates locating specific products using search criteria.

- Availability check: allows users to see if books are available for purchase.

- Shopping cart management: allows users to select books, adjust quantities, and remove items before confirming the order.

- Total calculation: calculates the total cost of selected products based on their prices and quantities.

- Order placement: allows users to finalize the purchase request for selected books.

- Order cancellation: provides the option to cancel orders based on their status and the bookstore's established policies.

- **Order tracking**: allows users to check the preparation and delivery status of a purchase.

- **Delivery process management:** facilitates updating the order status while it is in transit to the customer.

These functions outline the general capabilities planned for the product. Specific conditions and detailed behavior will be documented in the system requirements section.

2.3.User Characteristics

The system will be aimed primarily at three types of users, distinguished by their activities and needs within the platform.

User type	Description	Key needs
Customer	End-user who uses the platform to browse, purchase, and track books.	Register/Log in, browse catalog, search books, manage shopping cart, place and pay for orders, view order history, and check delivery status.
Admin	Authorized bookstore staff responsible for store administration and order processing.	Register/Log in, view orders, manage books, manage inventory, and manage delivery assignments.
Delivery Driver	Personnel responsible for transporting orders from the bookstore to the customer's location.	Log in, view assigned deliveries, inspect delivery addresses, and update delivery status during transit.

Users are expected to have basic knowledge of how to operate electronic devices and web applications. Therefore, the interface must be intuitive, accessible, and easy to use.

2.4. Restrictions

- **Role-Based Access Control:** Strict permission segregation between Customer, Admin, and Delivery Driver profiles.
- **Catalog & Inventory Accuracy:** Dependent on updates performed by authorized Admins.
- **Timely Delivery Updates:** Status updates rely on Delivery Drivers promptly logging shipping progress.

2.5. Assumptions and Dependencies

The bookstore will provide initial catalog, price, and stock data.

Customers and drivers have Internet-connected devices with standard web browsers.

Delivery Drivers update delivery statuses accurately in real-time.

2.6. Future Requirements

Automated SMS/Email notifications for order milestones.

Rating and review system for books and delivery services.

Advanced book recommendation engine.

3. Specific Requirements

3.1. External Interfaces

3.1.1. User Interfaces

- **SR-UI-01 (Responsive Design):** The web interface must be responsive, ensuring optimal display and usability across desktop, tablet, and mobile browsers.
- **SR-UI-02 (Language and Usability):** The user interface shall be presented in Spanish and adhere to basic accessibility and legibility guidelines (contrast-compliant typography and touch-friendly controls).
- **SR-UI-03 (Book Catalog):** The catalog interface shall display key product details for each book: title, author, cover image, price in USD, stock status, and an "Add to Cart" button.
- **SR-UI-04 (Order Tracking View):** The customer's order tracking screen shall display a progress timeline reflecting order statuses (*Confirmed*, *In Preparation*, *In Transit*, *Delivered*, *Canceled*) alongside a map view showing the delivery destination.

3.1.2. Hardware Interfaces

- **SR-HI-01:** The system requires no dedicated hardware; it operates on standard web servers and responds to client requests from internet-enabled consumer devices.

3.1.3. Software Interfaces

- **SR-SI-01 (Web Browsers):** The web platform must be compatible with modern major

web browsers, including Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.
- **SR-SI-02 (Mapping Service API):** The system shall integrate with an external mapping API (e.g., Google Maps or OpenStreetMap) to render delivery locations and visual progress tracking across Quito.

3.1.4. Communication Interfaces

- **SR-CI-01 (HTTPS Protocol):** All data communications between client browsers and the backend server must be encrypted using HTTPS with TLS 1.2 or higher.
- **SR-CI-02 (Data Exchange):** The client-server interactions shall rely on structured JSON payloads transmitted via RESTful APIs.

3.2. System Functions

3.2.1. Customer Functions

- **SR-FUN-CUST-01 (Register & Log In):** The system shall allow customers to create an account and log in securely.
- **SR-FUN-CUST-02 (Browse & Search Catalog):** The system shall allow customers to browse books, search by title/author, and view detailed book descriptions.
- **SR-FUN-CUST-03 (Shopping Cart Management):** Customers shall be able to add, adjust quantities, or remove books from their cart.
- **SR-FUN-CUST-04 (Order Placement & Payment):** Customers shall be able to place an order and complete payment for selected items.
- **SR-FUN-CUST-05 (View Orders & Check Delivery Status):** Customers shall be able to view their orders and check real-time delivery status.

3.2.2. Bookstore Staff Functions

- **SR-FUN-ADMIN-01 (Authentication):** Admins shall register and log in to access administrative features.
- **SR-FUN-ADMIN-02 (Manage Books & Inventory):** Admins shall add, edit, or remove book catalog entries and update inventory levels.
- **SR-FUN-ADMIN-03 (View Orders & Manage Deliveries):** Admins shall review customer orders and assign them to delivery drivers.

3.2.3. Delivery Person Functions

- **SR-FUN-DELI-01 (Authentication):** Delivery drivers shall log in to access assigned delivery duties.
- **SR-FUN-DELI-02 (View Assigned Deliveries & Addresses):** Drivers shall view their assigned delivery list and access delivery address details.
- **SR-FUN-DELI-03 (Update Delivery Status):** Drivers shall update delivery statuses (e.g., In Transit, Delivered) during order handoff.

3.2.4. System Maintenance & Backend Functions

- **SR-FUN-SYS-01 (Role-Based Access Control):** Enforce strict authentication and authorization across Customer, Admin, and Delivery Driver profiles.
- **SR-FUN-SYS-02 (Automatic Inventory Deduction):** Automatically reduce inventory upon order payment confirmation.

3.3. Performance Requirements

3.3.1. Response Time and Capacity

- **SR-PER-01 (Page Load Speed):** Catalog browsing and product pages shall load in under 2.5 seconds on standard broadband or cellular connections.
- **SR-PER-02 (Concurrent Users):** The platform shall support at least 100 concurrent browsing users and execute up to 15 order transactions per minute without service degradation.
- **SR-PER-03 (Search Performance):** Search queries against a database of up to 10,000 book records shall return results within 1.0 second.

3.3.2. Data and Synchronization

- **SR-PER-04 (Database Scalability):** The database shall accommodate an initial volume of at least 50,000 yearly orders and complete status logs.
- **SR-PER-05 (Status Synchronization):** Status updates recorded by staff or delivery personnel shall reflect on the customer's tracking view within 5 seconds.

3.4. Design Constraints

- **SR-CON-01 (Architecture):** The system shall follow a decoupled web architecture (separated Frontend interface and REST API Backend).
- **SR-CON-02 (Relational Database):** A relational database system (RDBMS, e.g., PostgreSQL or MySQL) must be used to ensure ACID compliance for transactions and inventory consistency.
- **SR-CON-03 (Web Standards):** All web development shall follow standard W3C guidelines for cross-browser compatibility.

3.5. System Attributes

- **SR-SEC-01 (Security & Encryption):** Salted password hashing (bcrypt/Argon2).
- **SR-AVA-01 (Availability):** 99% operational uptime during bookstore business hours.
- **SR-POR-01 (Portability):** Containerized backend deployment via Docker.

3.6. Other Requirements

- **SR-OTH-01 (Currency & Tax):** Calculations in USD with local Ecuadorian IVA tax.
- **SR-OTH-02 (Time Zone):** Order timestamps logged in Ecuador continental time (UTC-5).